DATA SCIENTIST | MACHINE LEARNING

FATIMA ADDA-REZIG

Yvelines (78) · Île-de-France · Available immediately · adda.fatima@hotmail.fr | GitHub | Portfolio

Biochemical engineer with 5 years of R&D experience in protein production, purification and characterization (academic laboratory and CRO), now a certified Data Scientist (French RNCP Level 7, Master's level). I design and deploy machine learning, deep learning and generative AI models (Python, PyTorch, MLflow, Docker, CI/CD, RAG).

Objective: I am looking for a Data Scientist position in biotech, pharma, CRO or medtech, where my understanding of experimental data speeds up R&D decisions.

PROFESSIONAL EXPERIENCE AND DATA SCIENCE PROJECTS

Data Scientist / Machine Learning | Applied training projects · OpenClassrooms | Mar – Dec 2026

Protein solubility — ML & Deep Learning — DeepSol data audit, homology control, LightGBM/ESM-2 comparison, fine-tuning, calibration and support for experimental decisions; Gradio demo online.

Technologies: Python, PyTorch, ESM-2, LightGBM, Optuna, MLflow, Gradio, Hugging Face · GitHub

RAG / LLM API — event search — Open data ingestion, FAISS indexing, semantic search, Mistral generation and source relevance control; API deployed on Render.

Technologies: Python, FastAPI, LangChain, FAISS, Mistral AI, Render · GitHub

Document assistant — OCR & LLM — OCR extraction, retrieval-augmented generation over documents and SQL data; faithfulness and answer relevance testing

Technologies: Nanonets OCR, Mistral, SQL, FAISS, RAGAS, Streamlit · GitHub

Brain MRI classification — Deep Learning — Prototyping on partially labeled images: CNN, transfer learning and semi-supervised approach.

Technologies: Python, PyTorch, CNN, OpenCV · GitHub

Research Associate in Biology | NOVALIX · CRO, Romainville | Oct 2024 – Jan 2026

Molecular Biology: Construct design, PCR, molecular cloning, DNA amplification: Maxi-Prep/Midi-Prep, transformation

Cell Culture: Development and adaptation of recombinant protein production protocols in various expression systems (E. coli, Sf9/Sf21 insect cells, HEK293-6E mammalian cells, ExpiCHO)

Protein Biochemistry: Purification of soluble and complex membrane proteins using HPLC systems; quality control of protein samples.

Structural Biology: Design of co-crystallization strategies and crystallogenesis according to client requests.

Research Engineer | I2BC · CNRS / CEA / Université Paris-Saclay | Dec 2017 – Feb 2021

- Research in molecular and structural biology; protein purification, protein/DNA interactions (SEC, switchSENSE, AUC), crystallography.
- Optimization of reproducible protocols, data analysis, documentation and scientific communication.

EDUCATION

Data Scientist – Machine Learning | OpenClassrooms · RNCP Level 7 (Master's level) · 2026 · 13 projects defended and validated

Data Scientist | DataScientest · 2021 · 400 h: Python, SQL, PySpark, ML, Deep Learning, NLP

Master's degree (M2) in Biochemistry, Molecular Biology and Biotechnology | Université d'Orléans · 2017

TECHNICAL SKILLS & LANGUAGES

Data Analysis — Python, pandas, NumPy, SQL / PostgreSQL, exploration, data cleaning and quality control.

Data Visualization — Matplotlib, Seaborn, dashboards, Streamlit, scientific reporting.

Data Science & AI — scikit-learn, LightGBM, XGBoost, PyTorch, CNN, Optuna, SHAP, LLM, RAG, OCR, LangChain, FAISS, RAGAS.

Data Drift & MLOps — Drift monitoring, MLflow, FastAPI, Docker, Git/GitHub, GitHub Actions, pytest, Hugging Face, Render.

Languages: English — C1 / Arabic — C1